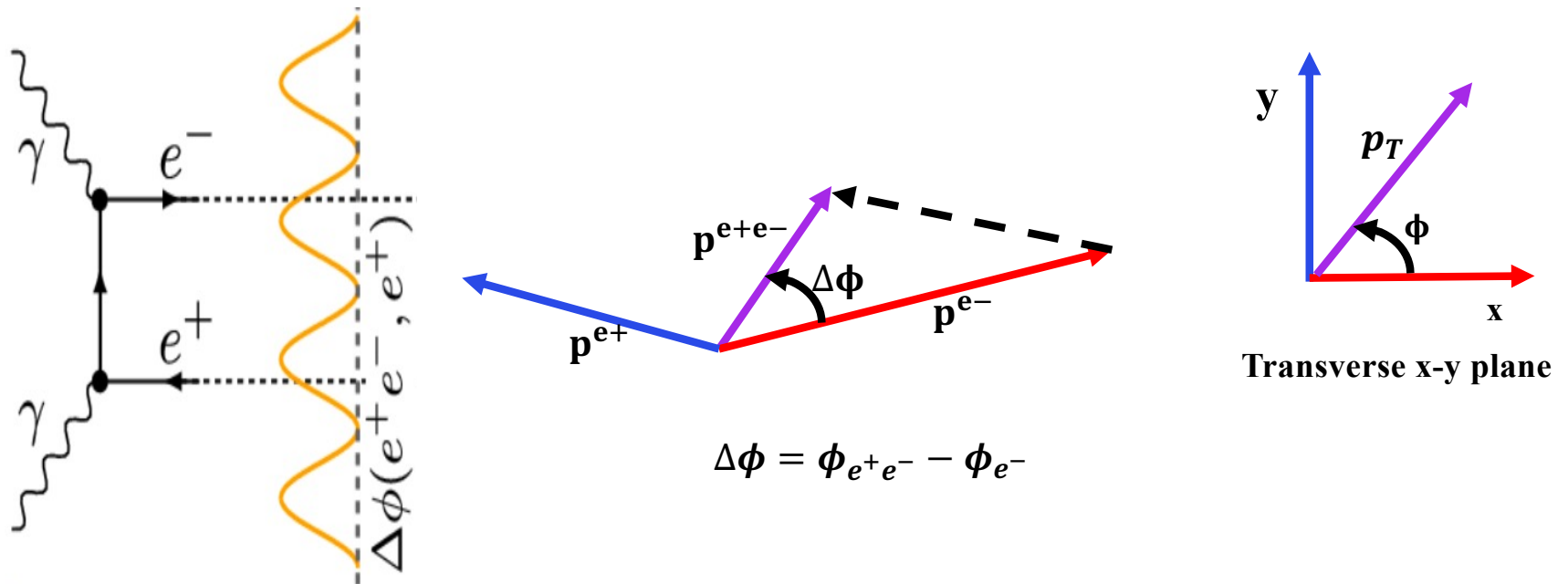
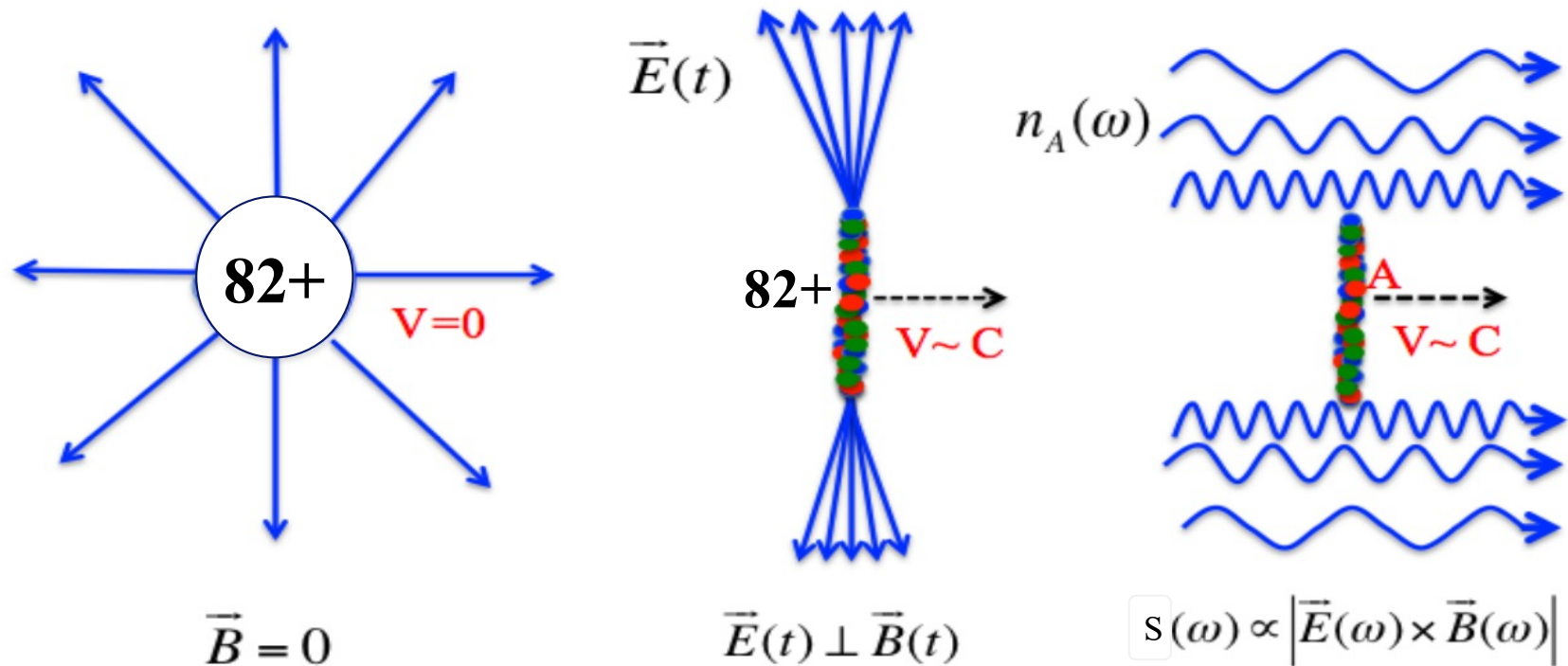

Measuring Angular Correlations in e^+e^- Pairs from Linearly-Polarized $\gamma\gamma$ Collisions



The Breit-Wheeler process generates matter and antimatter through photon-photon collisions

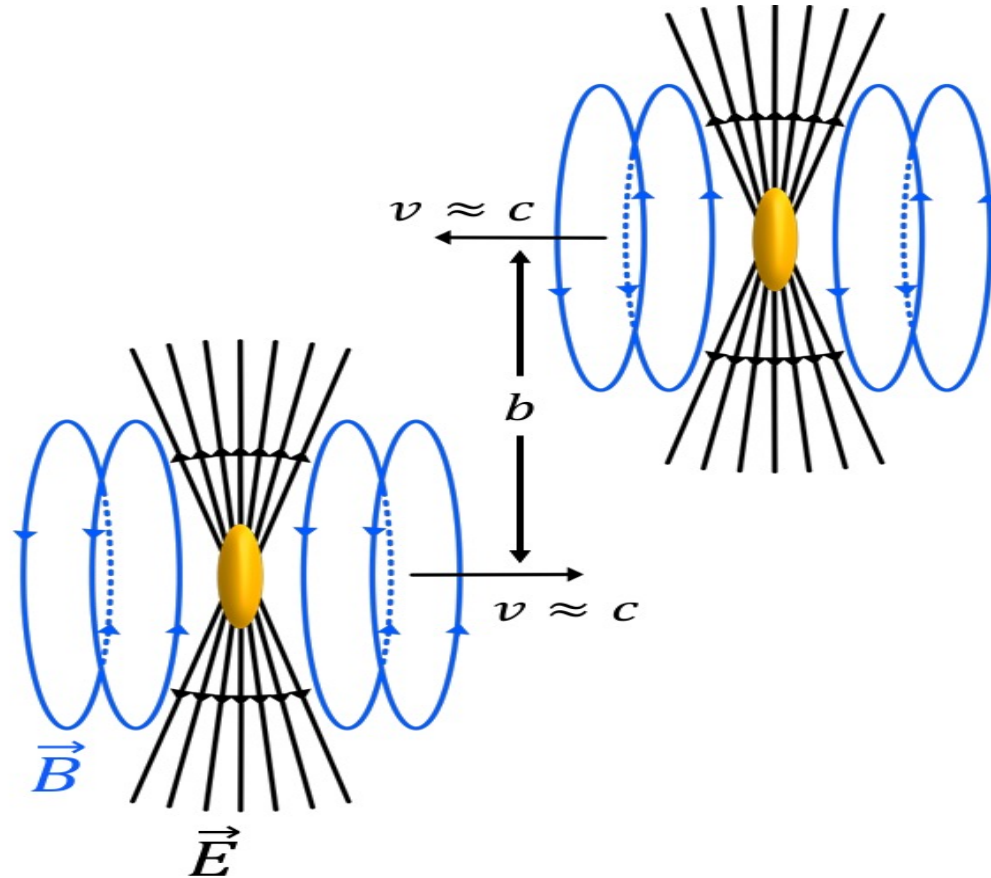
Electric and magnetic fields from a moving charge



Circular B field wraps around the direction of motion

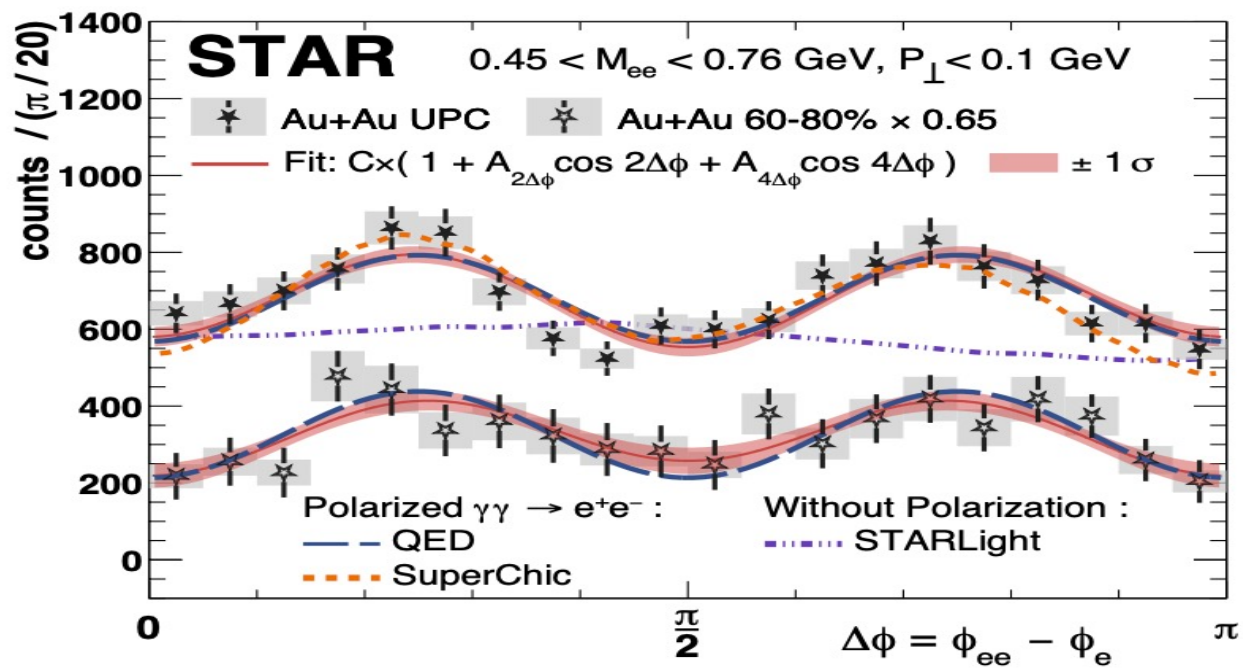
The photons are polarized because the field wraps around the z-axis

Ultra-relativistic charged nuclei produce highly Lorentz contracted electromagnetic field.



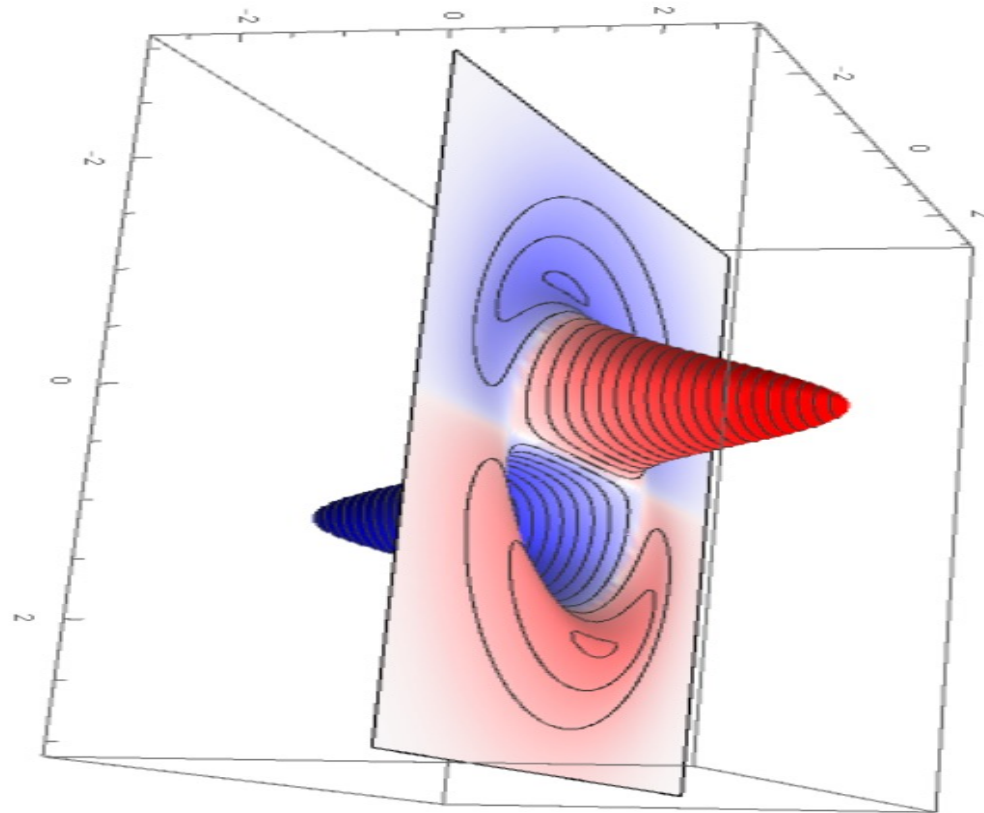
Previous Measurement from STAR

The measurement shows a significant fourth-order angular modulation of $\cos 4\Delta\phi = (16.8 \pm 2.5)\%$ and a smooth invariant mass distribution with no detectable vector mesons (ϕ , ω , ρ), with yields below the experimental limit of $\leq 0.2\%$.



Magnetic Field Near The Two Nuclei

magnetic fields of
order 10^{14} Tesla



Key points about this measurement

This process is essentially the creation of an electron-positron pair from two photons, known as the Breit-Wheeler process, which is only possible due to the strong electric fields associated with highly energetic photons. When heavy ions are accelerated at the LHC, they generate **magnetic fields of order 10^{14} Tesla**.

The key aspect is that the photons are linearly polarized, meaning their electric field oscillates along a specific direction, which in turn affects the angular distribution of the produced electron-positron pair.

Linearly polarized photon collisions are a **powerful tool** to probe the vacuum and a very strong magnetic fields from a two real photons

Compact Muon Solenoid

CMS DETECTOR

Total weight : 14,000 tonnes
Overall diameter : 15.0 m
Overall length : 28.7 m
Magnetic field : 3.8 T

STEEL RETURN YOKE
12,500 tonnes

SILICON TRACKERS
Pixel ($100 \times 150 \mu\text{m}$) $\sim 1\text{m}^2 \sim 66\text{M}$ channels
Microstrips ($80 \times 180 \mu\text{m}$) $\sim 200\text{m}^2 \sim 9.6\text{M}$ channels

SUPERCONDUCTING SOLENOID
Niobium titanium coil carrying $\sim 18,000\text{A}$

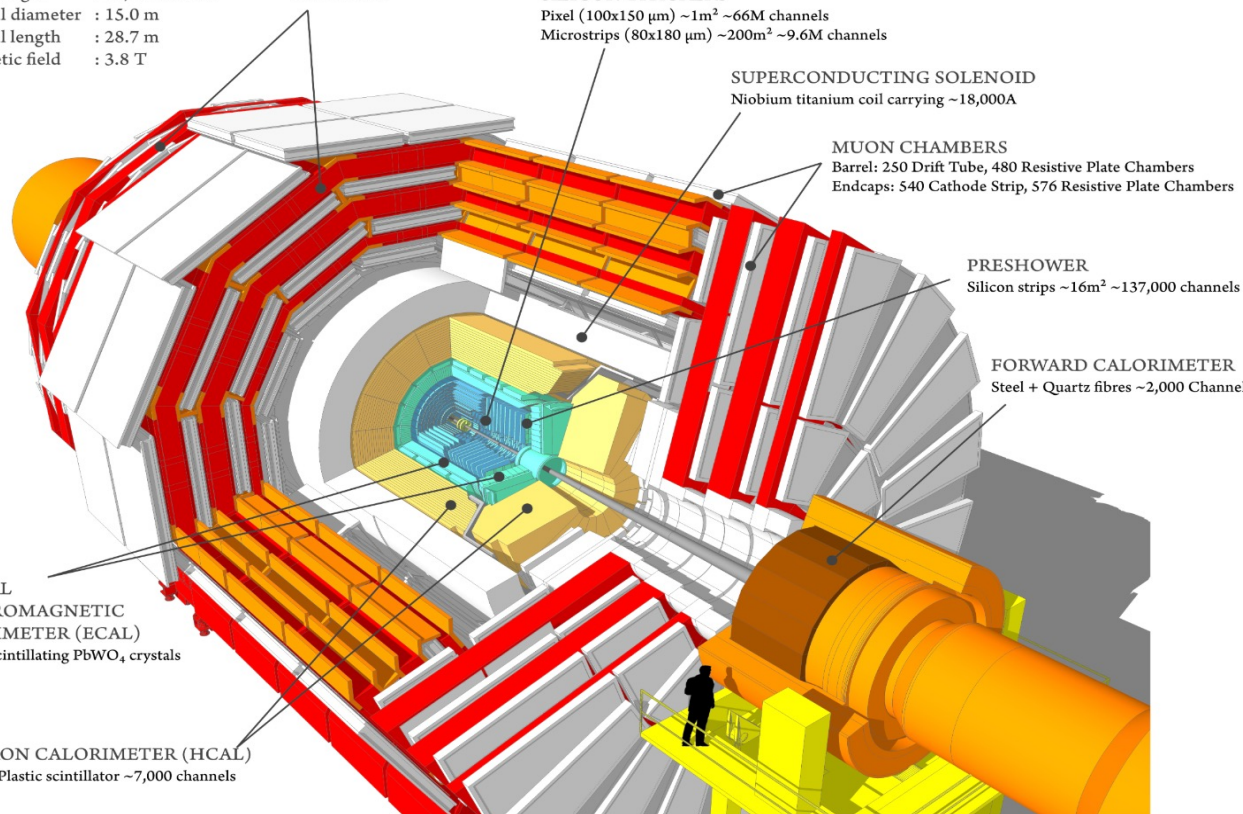
MUON CHAMBERS
Barrel: 250 Drift Tube, 480 Resistive Plate Chambers
Endcaps: 540 Cathode Strip, 576 Resistive Plate Chambers

PRESHOWER
Silicon strips $\sim 16\text{m}^2 \sim 137,000$ channels

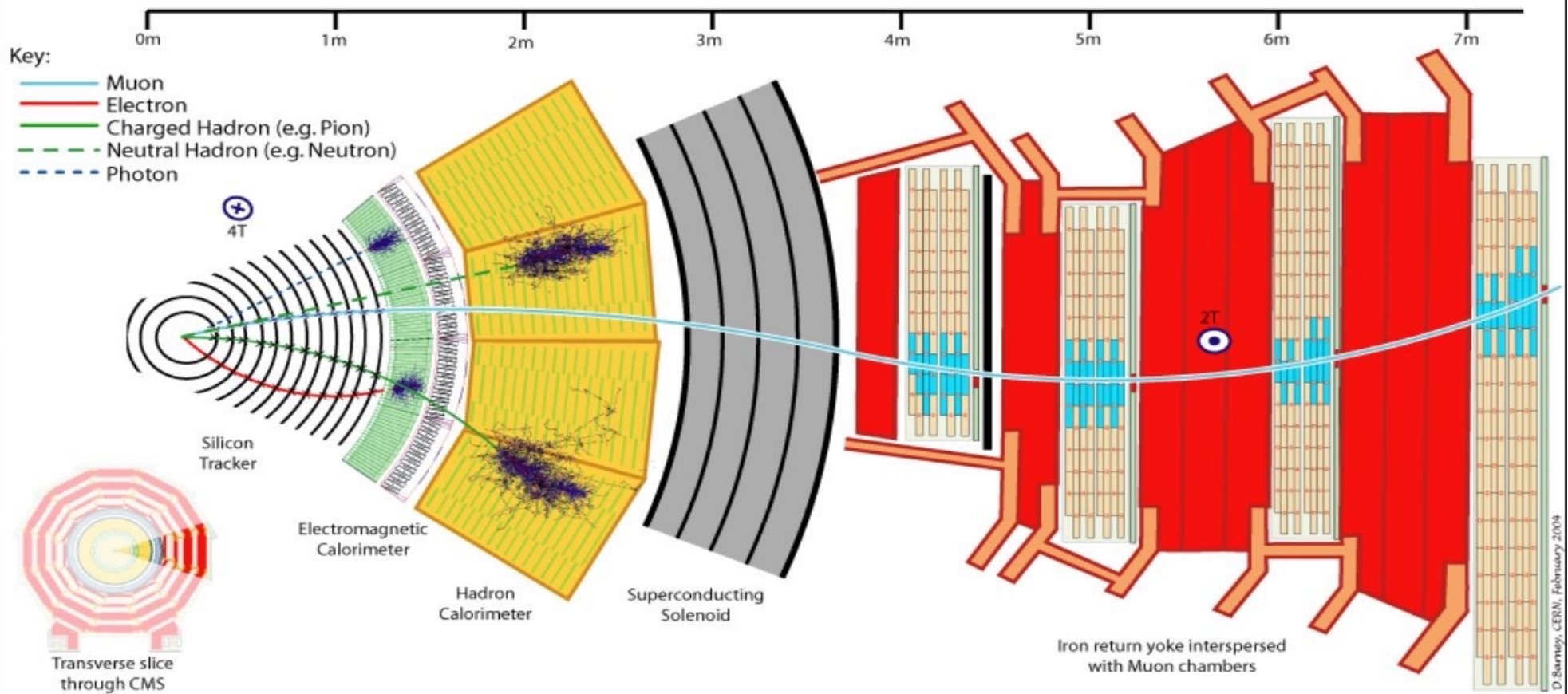
FORWARD CALORIMETER
Steel + Quartz fibres $\sim 2,000$ Channels

CRYSTAL
ELECTROMAGNETIC
CALORIMETER (ECAL)
 $\sim 76,000$ scintillating PbWO_4 crystals

HADRON CALORIMETER (HCAL)
Brass + Plastic scintillator $\sim 7,000$ channels



Transverse Slice Through CMS

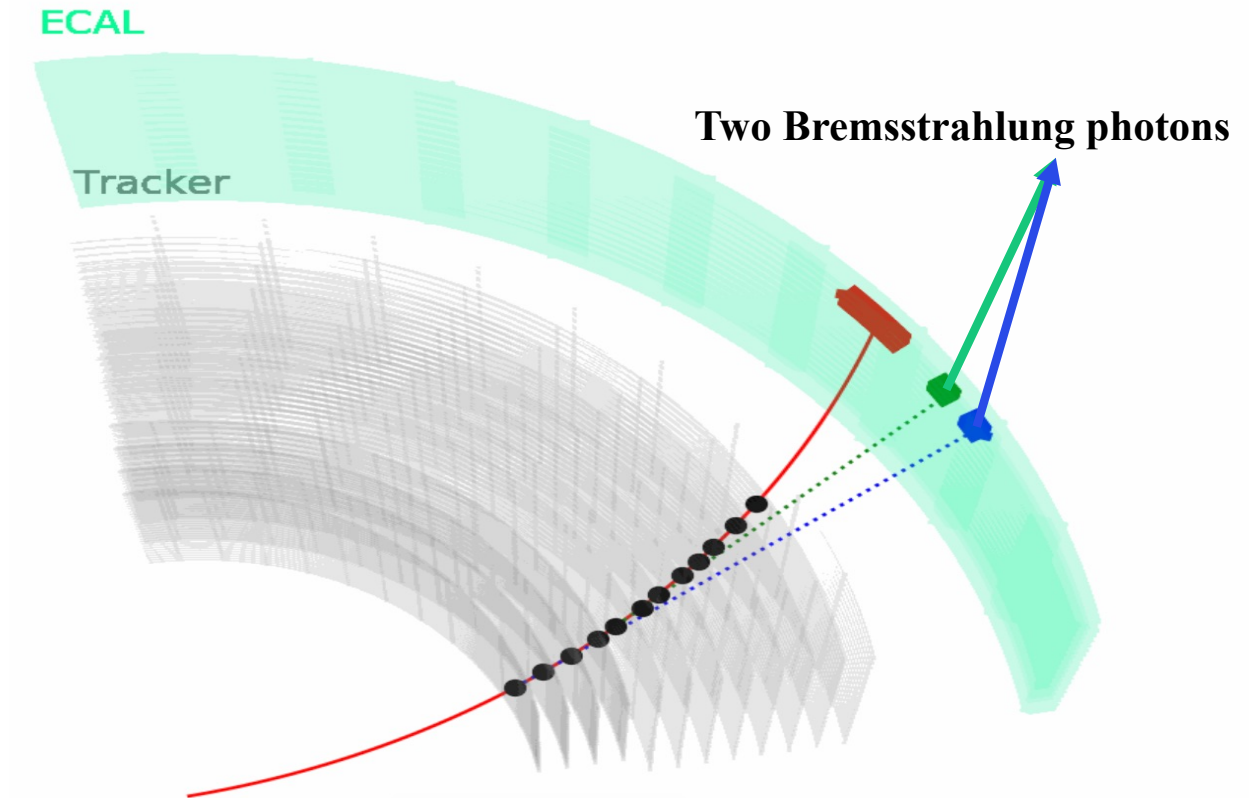


How is $\gamma\gamma \rightarrow e^+e^-$ Detected in CMS?

- ✓ **Silicon Tracker:** Both e^+ and e^- leave tracks due to their charge.
- ✓ **ECAL (Electromagnetic Calorimeter):** The electron and positron cause electromagnetic showers, depositing their energy.
- ✓ **Momentum Measurement:** The **curvature of their tracks** in the **magnetic field** helps measure their momenta and charge.
- ✓ **Signature in CMS:** Two oppositely charged tracks in the tracker + two electromagnetic showers in ECAL.

Identifying Electrons

$$R = \frac{p}{qB}$$

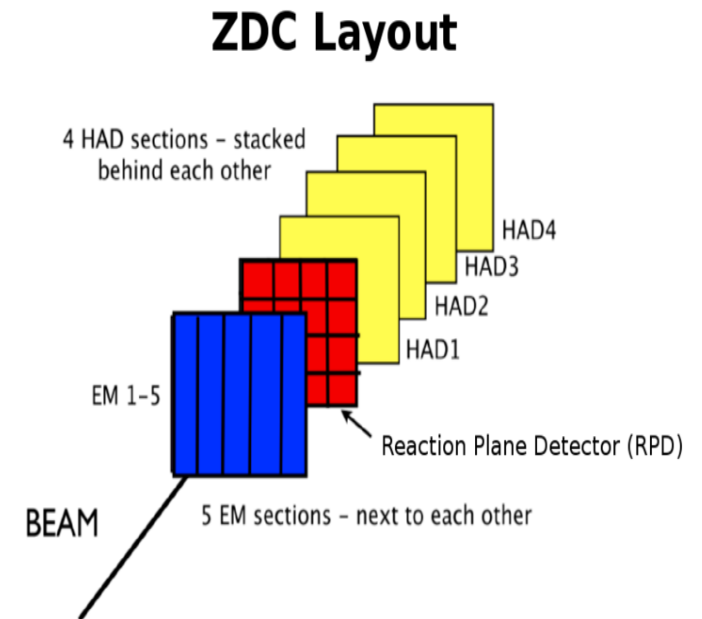
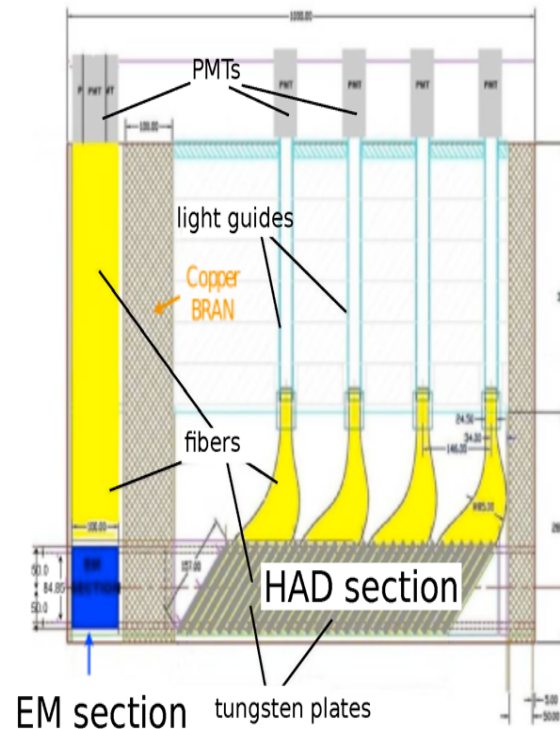


<https://cms.cern/news/finding-electrons-and-photons-cms-detector>

The Zero Degree Calorimeters (ZDCs)

The ZDC will be used to examine how neutron multiplicity affects mass and radius, which is an interesting area of study.

Photomultiplier Tubes (PMTs), which convert light signals into charge.

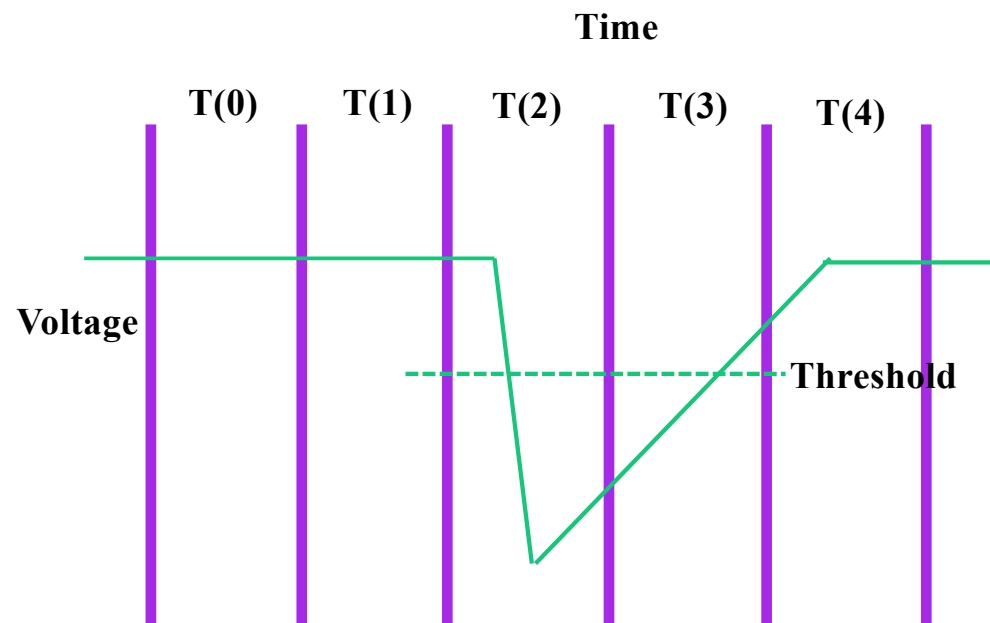
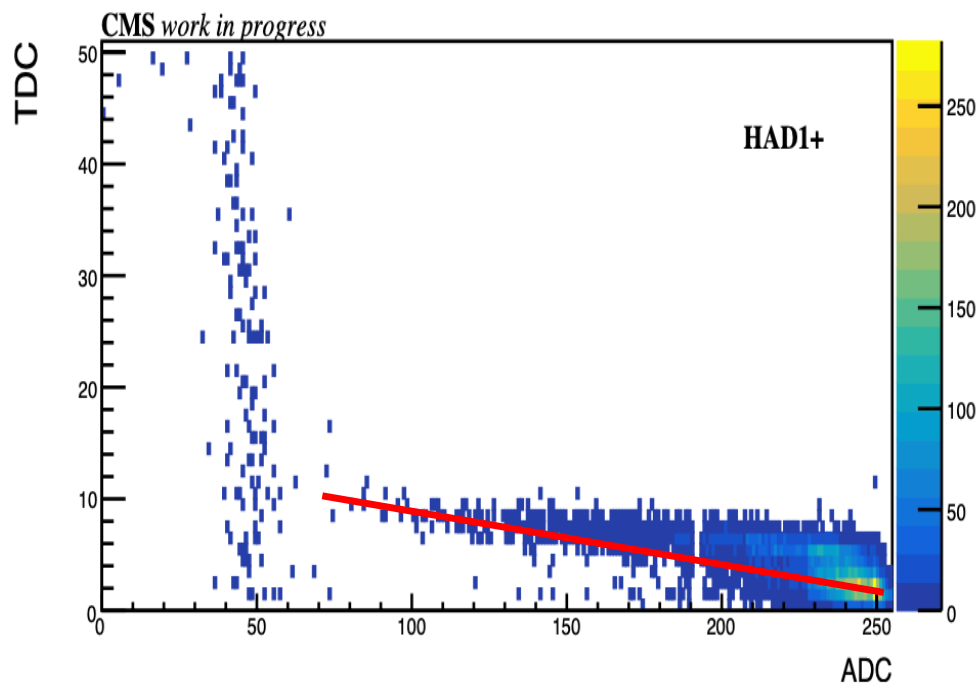


ZDC measure neutral particles, such as neutrons and photons, produced at very small angles (close to 0 degrees) relative to the beam direction.

Run 388095 PbPb 2024 Analysis in TS2

ADC: Amplitude Digital Converter

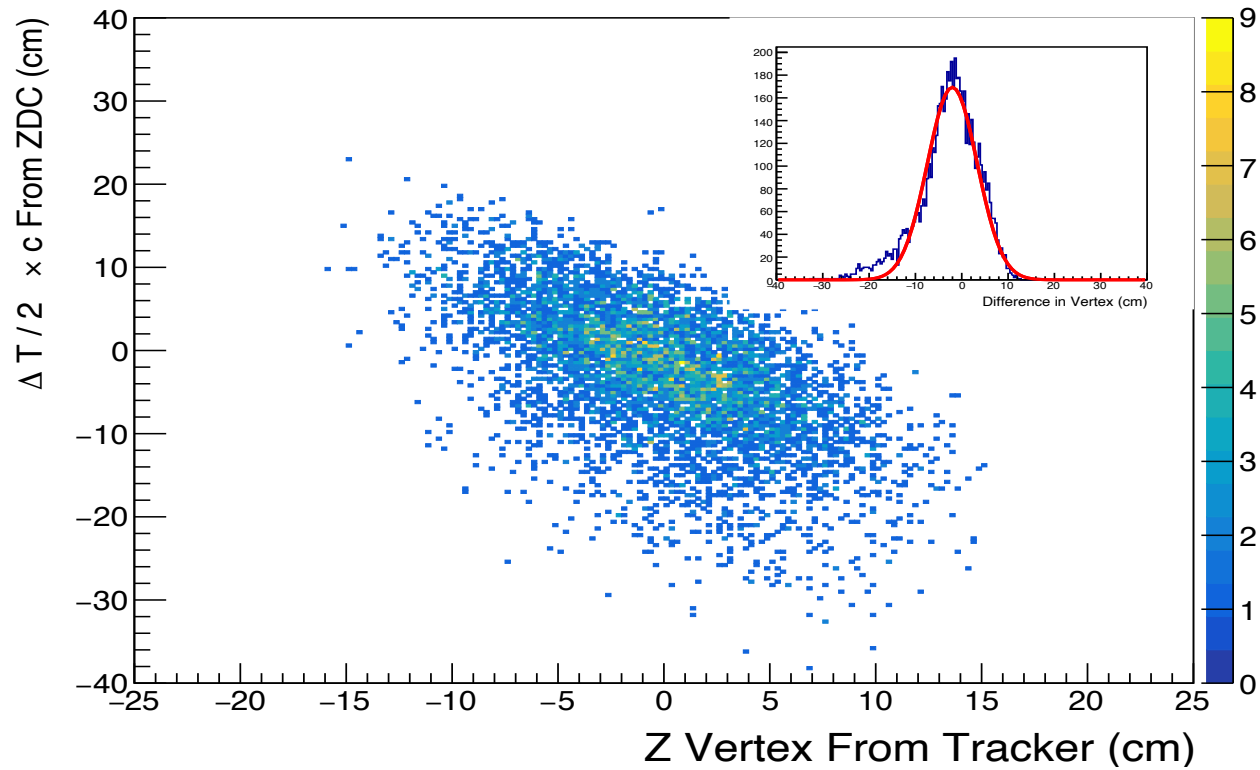
TDC: Time Digital Converter



The TDC records the precise moment when the green signal crosses the voltage threshold.

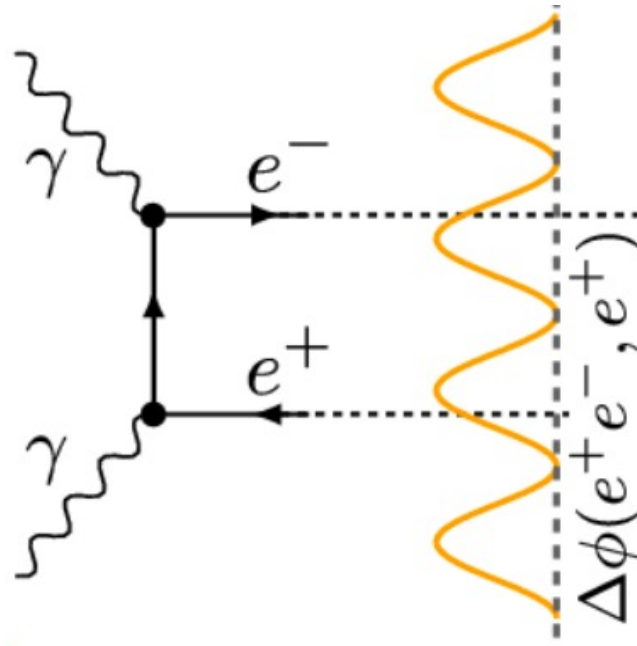
2024 PbPb: ZDC Vertex Resolution = 5.4cm

GOODNEWS!!!!!!



Proposal

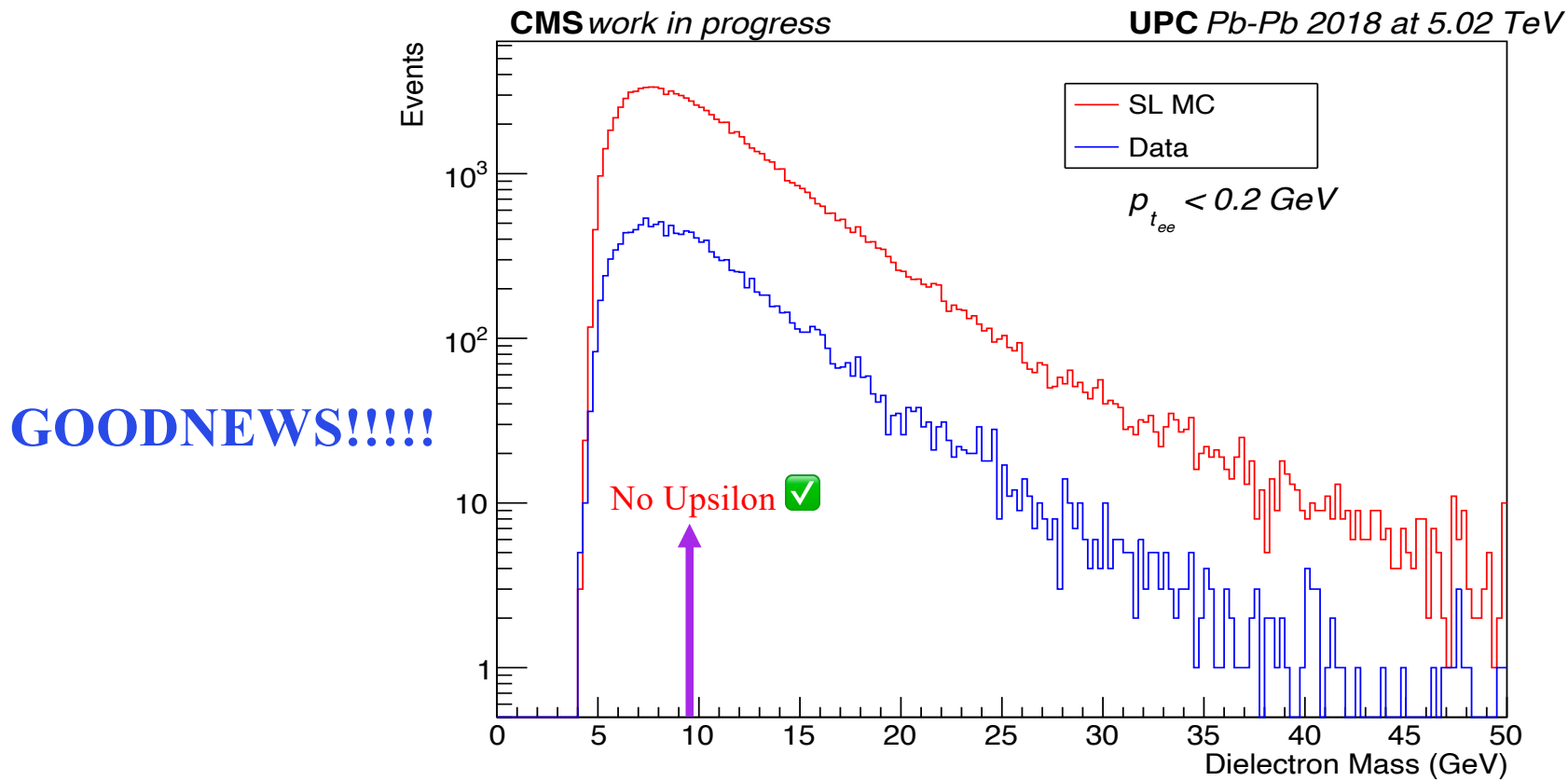
I propose to measure the angular correlations of electron and positron pairs in Pb-Pb UPC photon-photon collisions at $\sqrt{s_{NN}} = 5.36\text{TeV}$ within CMS. I will analyze 2023 and 2024 CMS data on electron-positron pair production in ultra-peripheral collisions, leveraging ~ 20 billion UPC events—far more than in 2018.



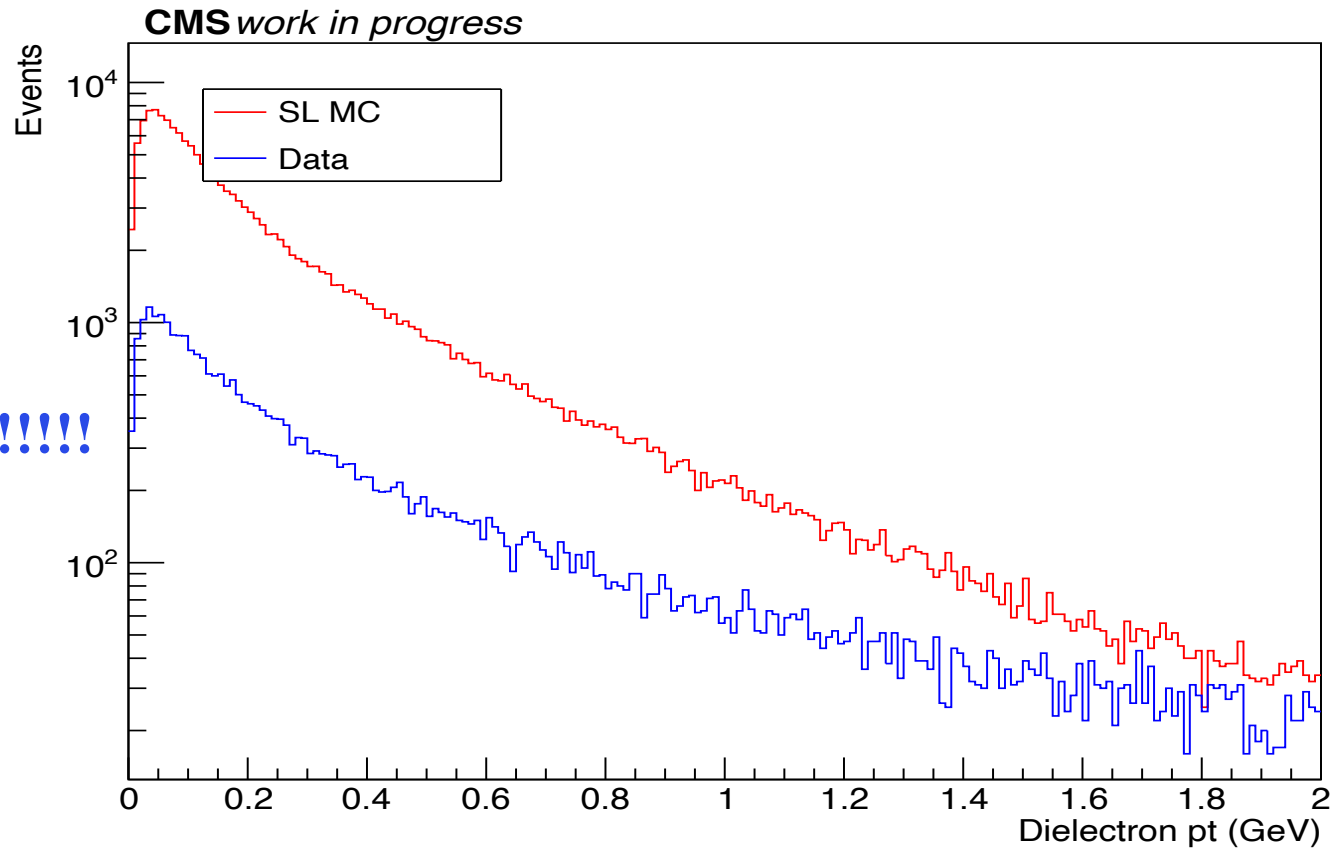
Outline for Data Analysis

1. **Electron Triggers:** The first part of the analysis will be devoted to identifying electron positron pairs. To select dielectron events, I will require that one of the UPC electron triggers fire, for example HLT HIUPC DoubleEG2 NotMBHF2AND v8.
2. **Selecting exclusive $\gamma\gamma \rightarrow e^+e^-$ events:** I expect $\gamma\gamma \rightarrow e^+e^-$ events to be back to back in the plane transverse to the beam momentum.
3. **Analysis of the Dielectron Kinematics:** I have looked at a small sample of data from the 2018 PbPb run that was produced for the light by light analysis within CMS.
4. **Correcting for the detector resolution:** It is essential to correct for distortions in the measured data caused by the finite resolution of the detector. To better understand the CMS response to PbPb collisions, I will unfold the signal to account for the effects of the detector's energy resolution using the RooUnfold package.
5. **Systematic Uncertainties:** I will investigate any systematic errors by analyzing how changes in the analysis affect the final results.

Mass Spectrum of Dielectron

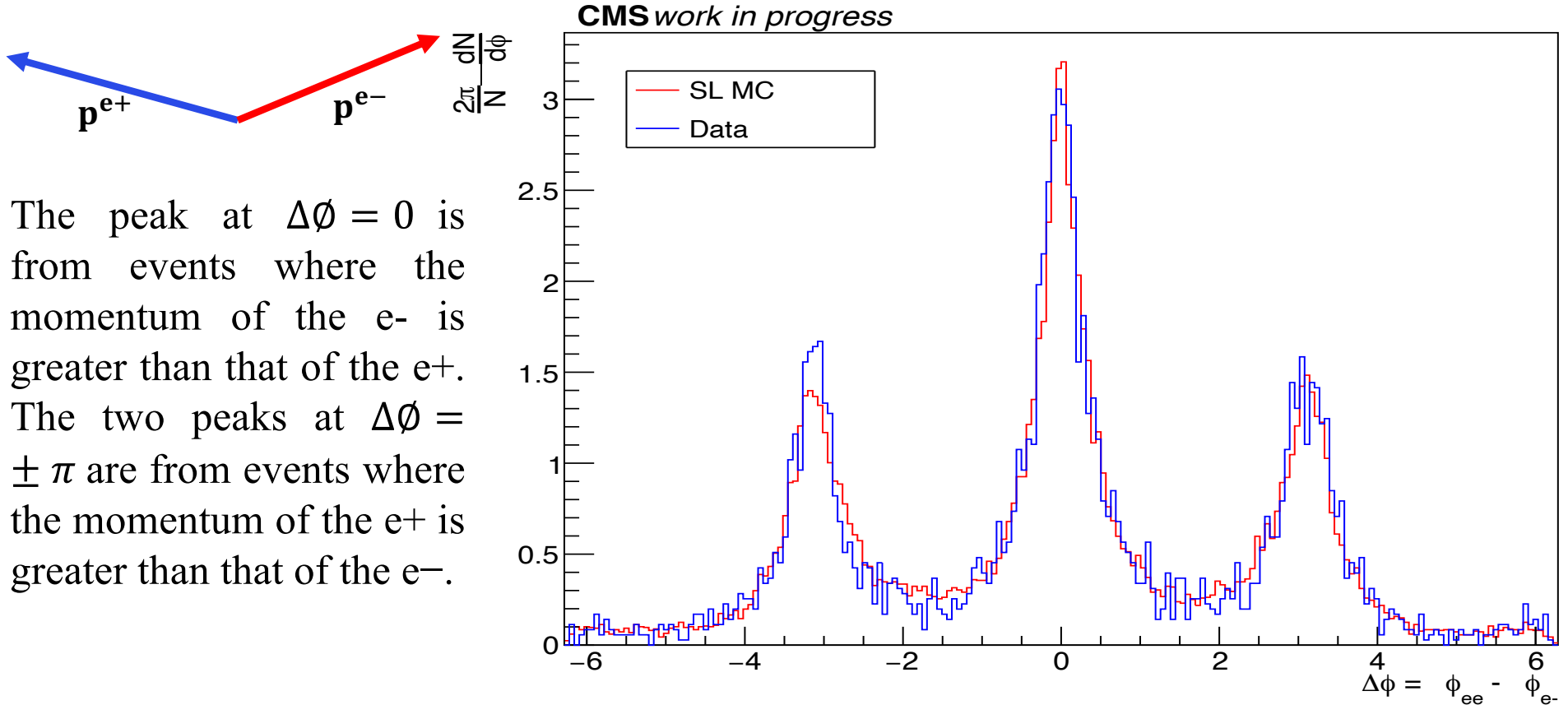


Momentum Spectrum of Dielectron



GOODNEWS!!!!!!

Analysis of the Dielectron Kinematics



Summary

- ✓ **High-Energy Angular Correlations:** This study aims to analyze electron-positron angular correlations in ultra-peripheral Pb-Pb collisions at $\sqrt{s_{NN}} = 5.36$ TeV, significantly extending previous measurements by STAR at lower energies.
 - ✓ **Enhanced Magnetic Field Effects:** The increased energy level is expected to result in stronger magnetic fields, providing a unique opportunity to study their influence on the Breit-Wheeler process and vacuum polarization.
 - ✓ **Utilization of CMS and ZDC Data:** The CMS experiment's high-precision detectors, along with the Zero Degree Calorimeter (ZDC), will be crucial in identifying forward neutrons and refining impact parameter-dependent analyses.
 - ✓ **Dielectron Event Selection & Systematics:** Rigorous selection criteria and systematic uncertainty assessments will ensure accurate extraction of angular correlations, improving precision compared to previous studies.
 - ✓ **Comparative Analysis & Future Impact:** The results will be compared with Monte Carlo simulations such as SuperChic and UPCgen, providing valuable insights into QED effects in high-energy photon-photon interactions. STARLight predicts an isotropic distribution without polarization effects, while SuperChic includes photon helicity dependence based on electromagnetic field orientation.
-

Thank You

Back up

Derivation of the Ultra-Strong Magnetic Field in Heavy-Ion Collisions

In the transverse direction, the electric field scales approximately as:

$E_{\perp} \approx \frac{Ze}{4\pi\epsilon_0 r'^2} \frac{1}{\gamma}$, where $\gamma = (1 - \beta^2)^{-1/2}$, For **LHC heavy-ion beams**, $\gamma \sim 3000$, so, the electric field is extremely strong in the transverse plane.

Magnetic Field from a Moving Charge: $B = \frac{1}{c^2} v \times E$

and considering that the nucleus moves along the z-axis, the electric field is primarily in the transverse direction (x,y-plane). Taking the cross-product:

$$B_y \approx \frac{v}{c^2} E_x$$

Since $v \approx c$ at ultra-relativistic speeds, $B_y \approx \frac{E_x}{c}$

Derivation of the Ultra-Strong Magnetic Field in Heavy-Ion Collisions

Using the transverse electric field expression:

$$B_{\perp} \approx \frac{1}{c} \frac{Ze}{4\pi\epsilon_0 r'^2} \frac{1}{\gamma}$$

Converting to SI units $\left(\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N.m}^2/\text{C}^2\right)$,

$$B_{\perp} \approx \frac{9 \times 10^9 \times Ze}{cr'^2 \gamma}$$

For lead ($Z = 82$) nuclei at the LHC, $\gamma \sim 3000$, $r' \sim 1 \text{ fm} = 10^{-15} \text{ m}$

Plugging in values: $B_{\perp} \approx \frac{9 \times 10^9 \times 82e}{(3 \times 10^8)(10^{-30} \times 3000)}$

Approximating, $B_{\perp} \approx 10^{14} \text{ Tesla}$. The **strongest known magnetic fields in the universe**.

Ultra-Strong Magnetic Field in Heavy-Ion Collisions

Vacuum birefringence: Modifications of photon propagation in strong fields.

EPA (Equivalent Photon Approximation) is a method used in high-energy physics to describe electromagnetic interactions in terms of an effective flux of virtual photons. It simplifies complex calculations involving photon-mediated processes by treating them as if real photons were involved.

Vacuum birefringence is a quantum electrodynamical (QED) effect predicted by **Heisenberg and Euler** in 1936. It arises due to the interaction of photons with a strong external magnetic field, even though photons do not directly interact in classical electrodynamics.

Ultra-relativistic charged nuclei produce highly Lorentz contracted electromagnetic field. In a specific phase space, EM fields can be quantized as a flux of real photons

At $\Delta\phi=0$, what kind of occurrence will give me a peak?

A **dielectron event** refers to the production of an **electron-positron (e^+e^-) pair** in a high-energy collision. If a peak appears at $\Delta\phi=0$ for dielectron events, it means that the two electrons are emitted in the **same direction in the azimuthal plane**. This situation can arise in specific physical processes as follows:

Conversions of Photons to Electron-Positron Pairs ($\gamma\gamma \rightarrow e^+e^-$): A high-energy photon can interact with a nucleus (or detector material) and convert into an electron-positron pair. **Why $\Delta\phi=0$ peak?** The electron and positron are often emitted at very small opening angles, leading to nearly identical azimuthal angles (ϕ), and thus a peak at $\Delta\phi=0$.

Bremsstrahlung Followed by Pair Production: An electron emits a bremsstrahlung photon, which subsequently converts into an **e^+e^- pair**. **Why $\Delta\phi=0$ peak?** The kinematics of the process can lead to cases where the original electron and the newly produced pair travel in the same direction.

What Are Virtual Photons?

Virtual photons are **not real photons** in the sense that they do not satisfy the classical relation:

$$E^2 = p^2 c^2$$

Instead, they have a **nonzero invariant mass** and exist only as intermediaries in interactions. They appear in **quantum field theory (QFT)** as exchanged particles in Feynman diagrams.

- ✓ **Real photons:** Massless, travel at the speed of light.
- ✓ **Virtual photons:** Can have an effective mass, do not propagate freely, and exist only temporarily.

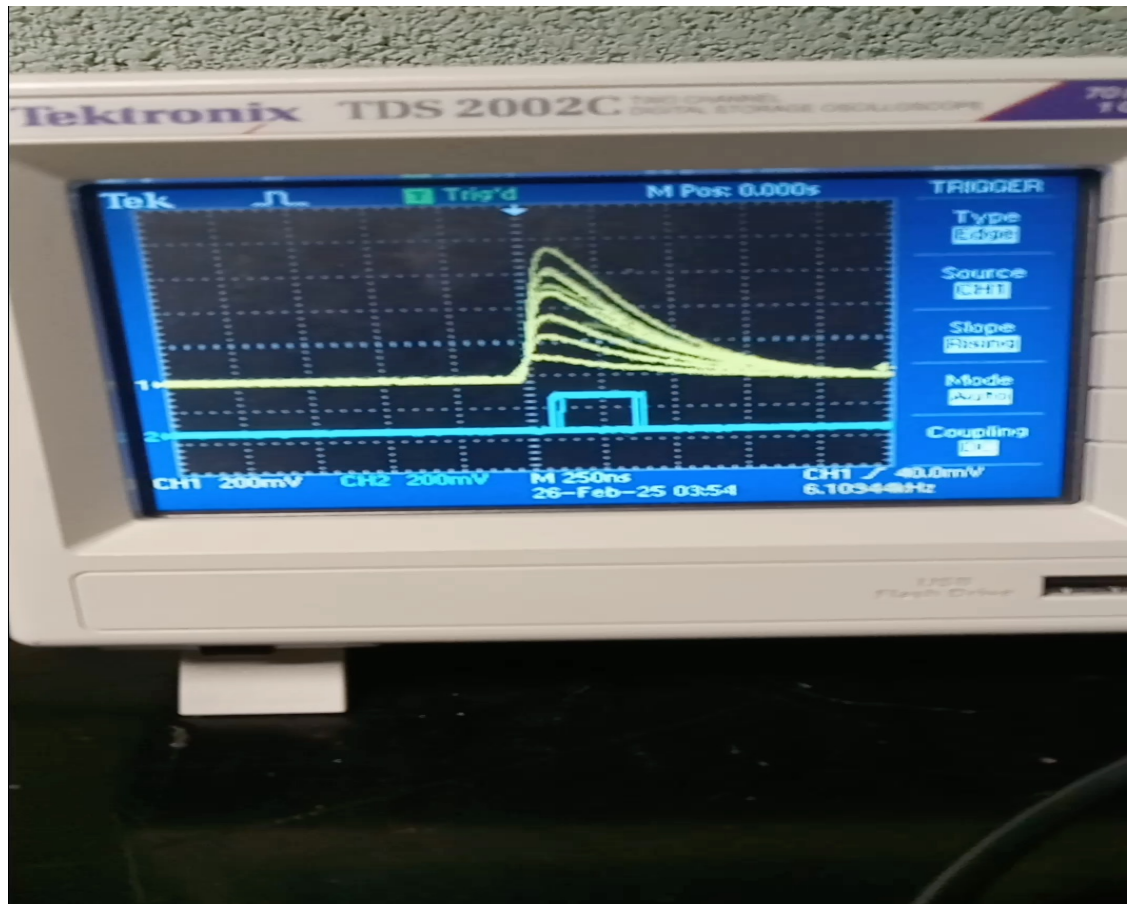
What is Exclusive e^+e^- Production?

Exclusive production of electron-positron (e^+e^-) pairs refers to a process in which **only** the final-state particles (e^+e^-) are observed, with no additional particle production. This happens via photon-photon interactions in (UPCs): $\gamma\gamma \rightarrow e^+e^-$. Unlike inelastic interactions, exclusive production occurs **without any fragmentation or hadronization**, meaning the colliding heavy ions remain intact or only slightly excited.

How Does $\gamma\gamma \rightarrow e^+e^-$ Occur in UPCs?

- ✓ **Heavy Ions as Photon Sources:** Heavy nuclei (such as lead, Pb) generate intense electromagnetic fields, which can be described as virtual photon clouds. **$\gamma\gamma$ interactions:** Two virtual photons interact, producing a final state
- ✓ **Photon-Photon Interaction:** When two heavy ions pass each other at distances greater than their nuclear radii, their virtual photons interact, producing an electron-positron pair: $Pb + Pb \rightarrow Pb + e^+e^- + Pb$
- ✓ **No Strong Interaction:** The ions **do not collide directly**, preventing any nuclear breakup.

TDC vs. ADC in TS2 Analysis



TDC vs. ADC in TS2

A **TDC (Time-to-Digital Converter)** is a device that converts a time interval (the time between two events) into a digital number. It measures the **time** between when a particle or signal passes through a detector and when it is recorded by the system.

An **ADC (Analog-to-Digital Converter)** converts an **analog signal** (continuous in time and amplitude) into a **digital signal** (discrete in both time and amplitude). The ADC records the **amplitude** of a signal at a particular point in time, usually as a digital value corresponding to the voltage level.

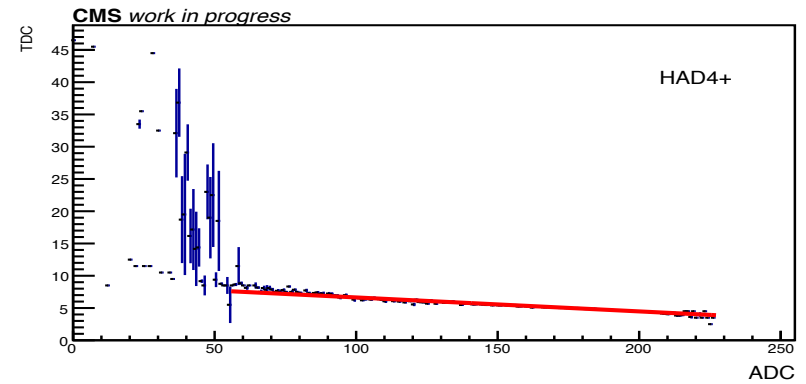
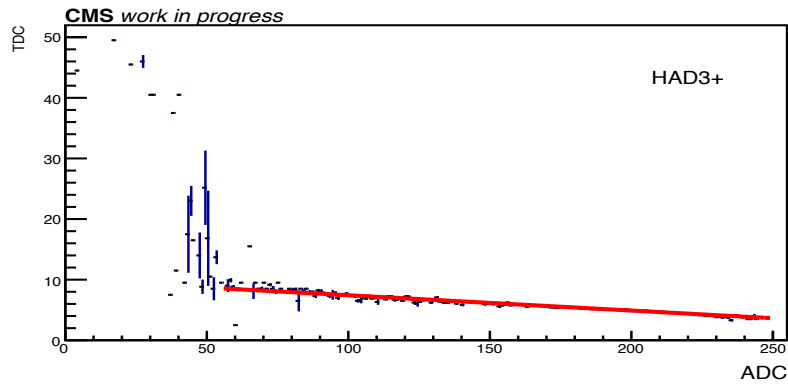
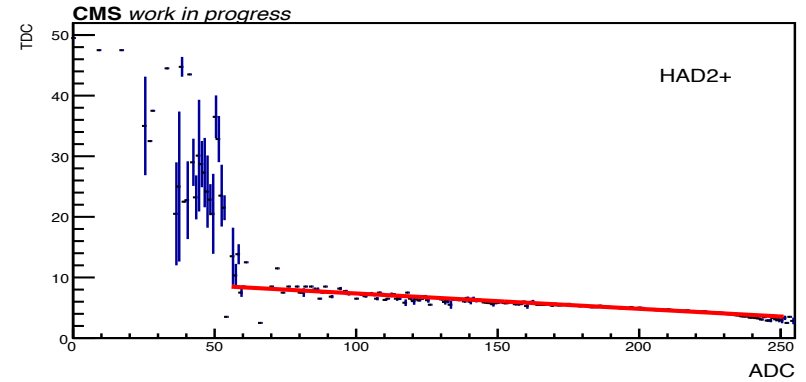
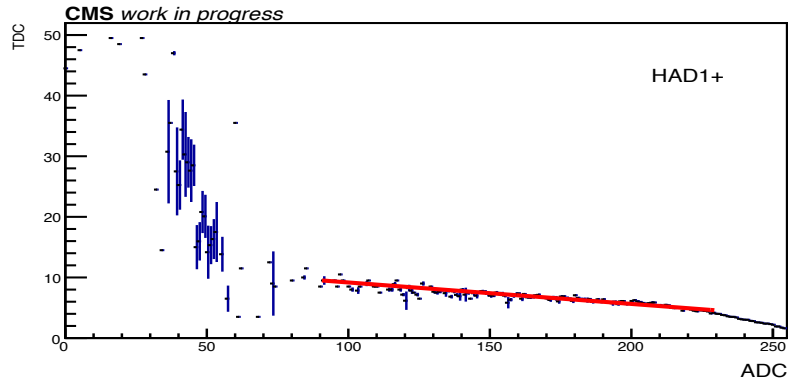
Every 25ns the ZDC electronics measures both the total charge in femto coulombs, and if the signal exceeds a certain threshold, the time in units of 0.5ns. Only signals above the threshold are considered meaningful events and are recorded by the system.

TDC: Measures **time intervals** (how long between two events).

ADC: Measures **signal amplitude** (energy or voltage level) and converts it to digital form.

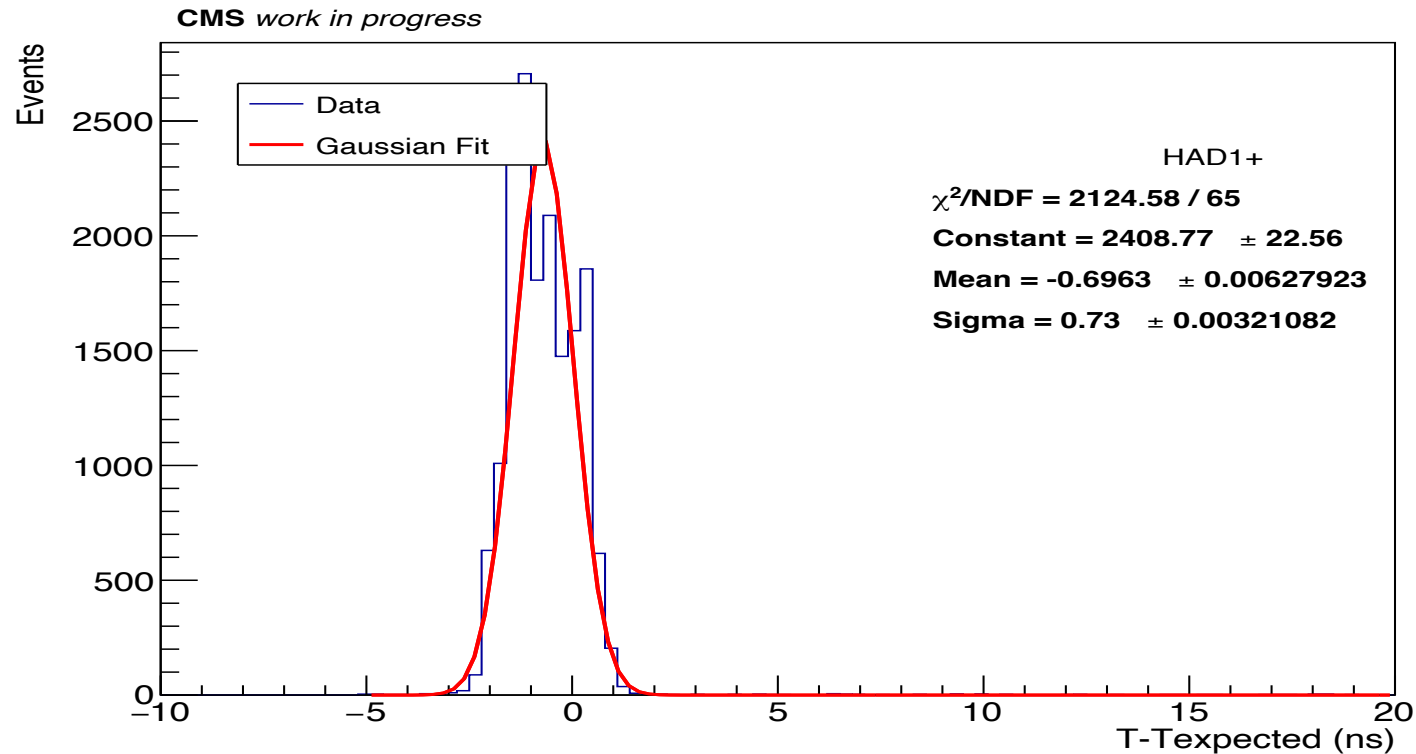
Events that lead to a peak at $\Delta\phi=0$ typically involve **back-to-back emissions** of particles, such as in **dijet production**, **lepton pair production**, or **electron-positron pair production**. These situations occur when particles are emitted in the same direction, creating a small azimuthal angle difference.

TDC vs. ADC in TS2



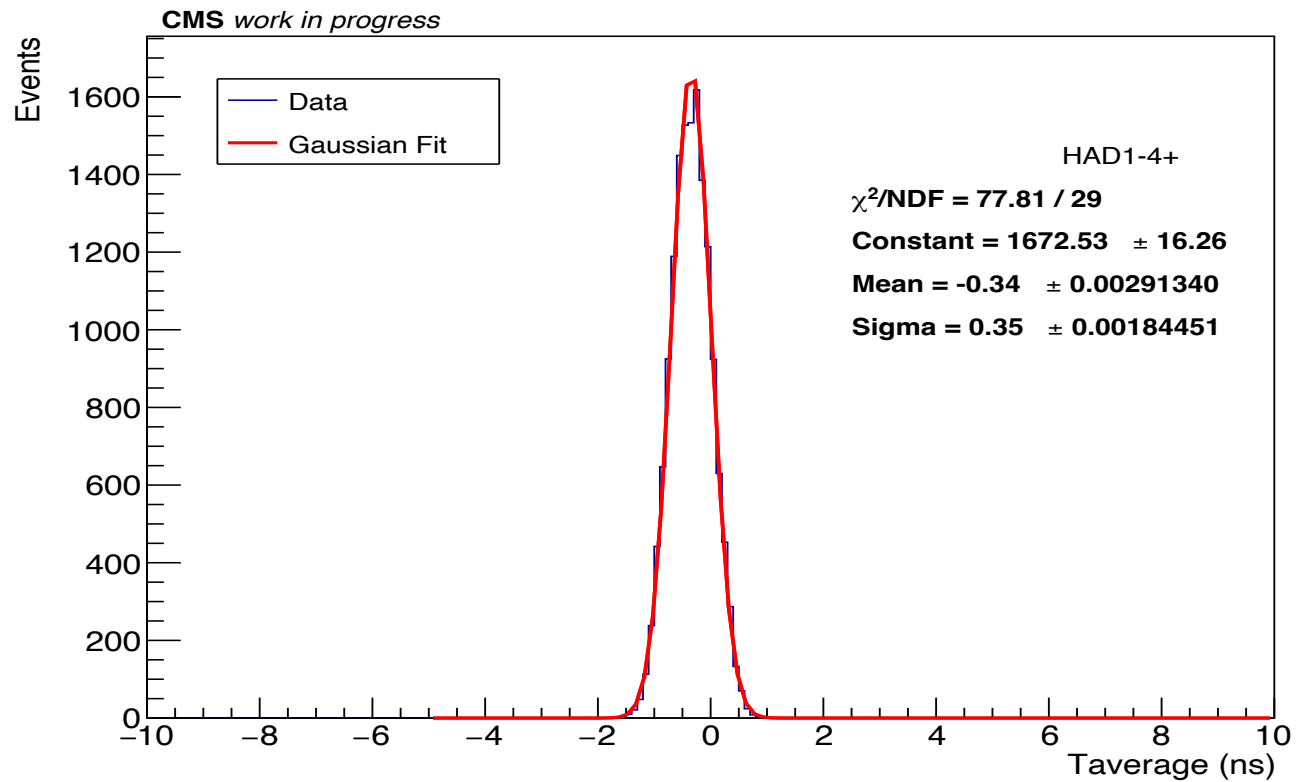
The fitted line is the signal. Here I considered only events that satisfy the TDC cut (TDC 60. Run 388095, PbPb 2024.

TDC vs. ADC in TS2



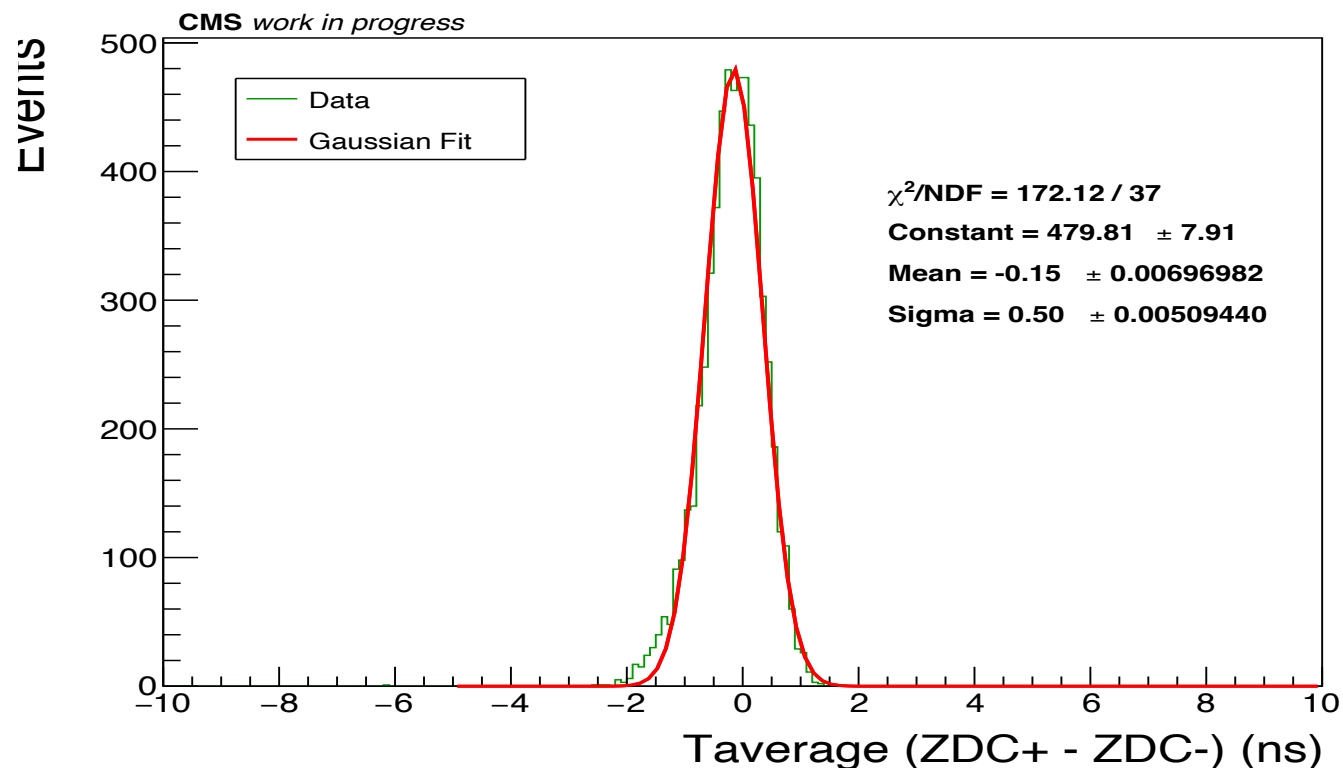
The quantity $T - T_{\text{expected}}$ in these plots represents the difference between the measured time (T) from the ZDC+ and the expected time (T_{expected}) based on a linear fit model. Run 388095 PbPb 2024.

TDC vs. ADC in TS2



Events with TDC cut < 51 show a 350 ps resolution.

TDC vs. ADC in TS2



ZDC+ Timing Performance: Taverage represents timing deviations in HAD1-4, comparing tdcTs2 with Texpected.

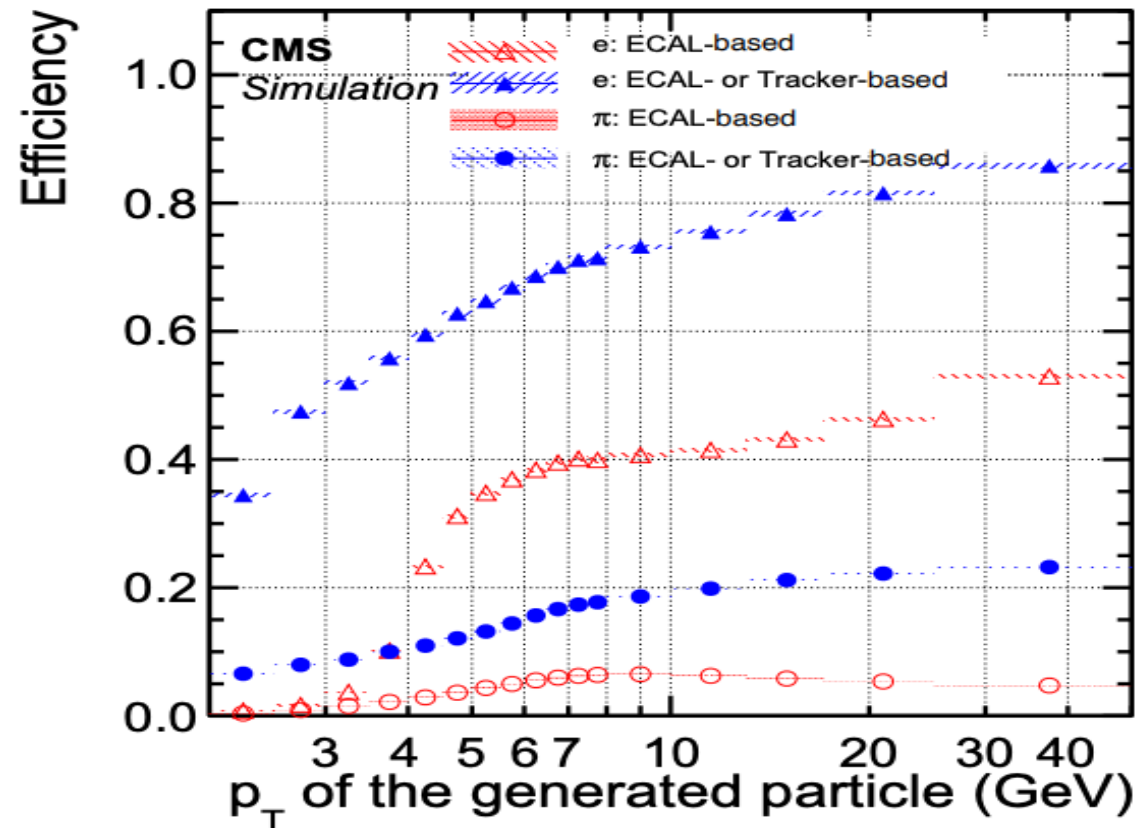
Ultra-Strong Magnetic Field in Heavy-Ion Collisions

The curvature of the trajectory gives a measurement of the particle's momentum. When the electron ends up interacting with the detector material, it will emit photons, which take away part of its energy and change its direction. This process is known as bremsstrahlung.

Particle Flow Algorithm in CMS and Electron Reconstruction

Observations

- ✓ Electrons (e) have higher efficiency than pions (π).
- ✓ ECAL-based reconstruction performs better than Tracker-based.
- ✓ Efficiency increases with p_T for both electrons and pions.
- ✓ Pions have much lower efficiency, particularly at lower p_T .
- ✓ Around 10 GeV and above, electron efficiency stabilizes close to 1.



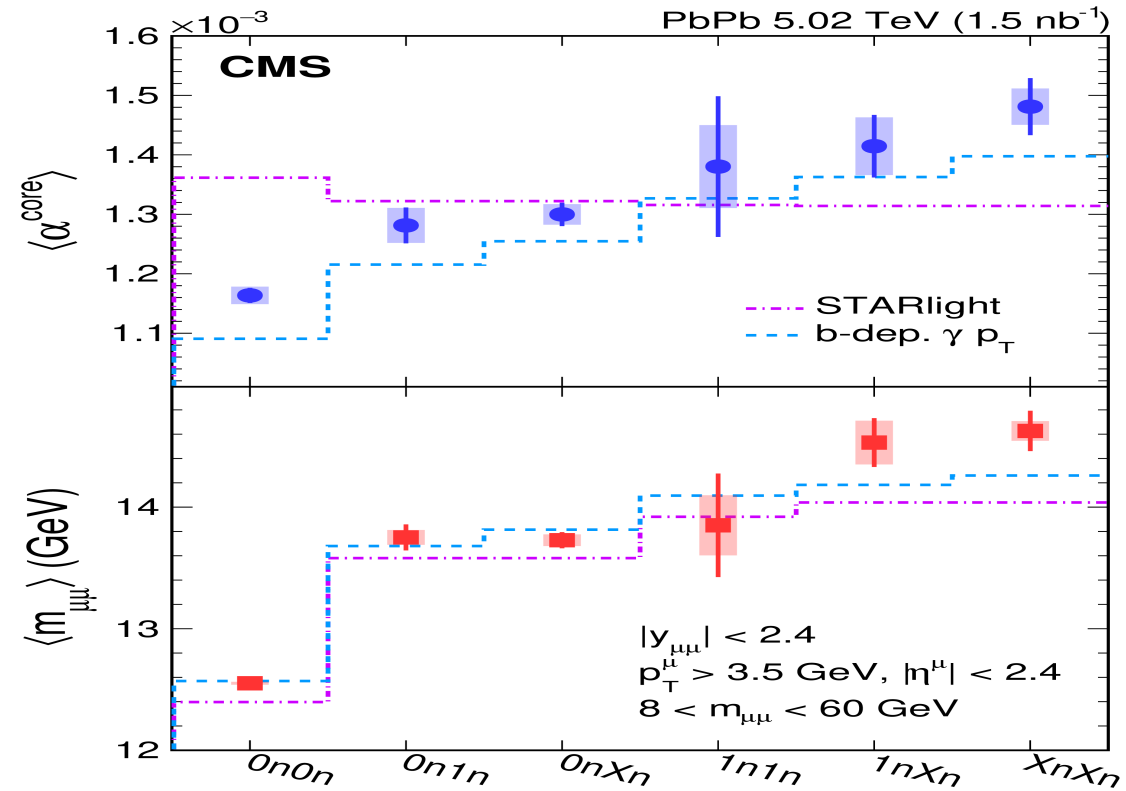
Ultra-Peripheral Collisions

The **Breit-Wheeler process** is the direct conversion of two photons into an electron-positron pair where as

Light-by-light (LbyL) scattering is the rare process where two photons scatter off each other

Vacuum Polarization: In extreme electromagnetic fields, the vacuum itself can become polarized, and the usual predictions of QED are altered. This results in an even more complex interaction between photons and produced particles.

Dimuons vs ZDC Multiplicity



Relativistic Mechanics

$$E^2 = p^2 c^2 + m^2 c^4$$

For an electron with:

Momentum $p = 1 \text{ GeV}/c$

Mass $m = 0.1 \text{ GeV}/c^2$

Speed of light $c = 3 \times 10^8 \text{ m/s}$

First, let's compute the total energy:

$$E = \sqrt{(1.0)^2 + (0.1)^2} \text{ GeV}$$

$$E = \sqrt{1.01} \approx 1.005 \text{ GeV}$$

Let's compute the Velocity v :

$$v = \frac{pc^2}{Ec} = \frac{pc}{E} = \frac{(1.0)c}{1.005} = 0.995c$$

$$v_e \approx 0.995c$$

Muon Lifetime and CMS Traversal Time

The **proper lifetime** of a muon at rest is $\tau_0 = 2.2 \text{ microseconds } (\mu s)$

However, due to **time dilation** at high energies, the observed lifetime increases:

$$\tau = \gamma \tau_0$$

Where γ is the Lorentz factor:

$$\gamma = \frac{E}{m}$$

For a **10 GeV muon** ($m=0.105 \text{ GeV}$):

$$\gamma = \frac{10}{0.105} \approx 95$$

So, the **dilated lifetime** is:

$$\tau \approx 95 \times 2.2 \mu s \approx 209 \mu s$$

Muon Lifetime and CMS Traversal Time

Time to traverse CMS:

A muon traveling at nearly **c** crosses the **CMS tracker** (**$\sim 1.3m$ radius**) **in:**

$$t = \frac{1.3m}{3.00 \times 10^8 \text{ m/s}} \approx 4.3ns$$

Since $209\mu s \gg 4.3ns$ the probability of a muon decaying to e^+e^- inside the tracker is **extremely low**.

Muons typically **decay far outside** CMS, in the **muon chambers** or even beyond.

Maxwell's Equation

Gauss's Law for Electric Fields:

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

Physical Meaning: The total electric flux through a closed surface is proportional to the charge enclosed within that surface.

Maxwell's Equation

Gauss's Law for Magnetism:

$$\vec{\nabla} \cdot \vec{B} = 0$$

Explanation: The magnetic field $\vec{B}$ has no beginning or end; it always forms continuous loops. The divergence of the magnetic field is zero, indicating there are no isolated magnetic poles (unlike electric charges).

Physical Meaning: The magnetic flux through any closed surface is zero.

Maxwell's Equation

Faraday's Law of Induction:

$$\vec{\nabla} \cdot \vec{E} = - \frac{\partial \vec{B}}{\partial t}$$

Explanation: A time-varying magnetic field induces an electric field. The curl of the electric field at a point is equal to the negative rate of change of the magnetic field.

Physical Meaning: A changing magnetic field produces a circulating electric field. This phenomenon is the basis for electromagnetism in alternating current (AC) circuits.

Maxwell's Equation

Ampère's Law (with Maxwell's correction):

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

Explanation: The curl of the magnetic field at a point is proportional to the electric current density $\vec{J}$ at that point and the rate of change of the electric field. μ_0 is the permeability of free space, a constant that quantifies the ability of a vacuum to carry a magnetic field.

Physical Meaning: Electric currents (moving charges) generate magnetic fields, and a time-varying electric field also generates a magnetic field.